

PROCESS INJECTION ATTACKS AND DETECTION

CHEAT SHEET

Sample Program Locations

Path	Description
<code>C:\injection\local\local_shellcode_inj.exe</code>	Local Injection
<code>C:\injection\htbdll.dll</code>	Sample HTB DLL
<code>C:\injection\remote\injection1.exe</code>	Remote Injection
<code>C:\injection\threadhijacking\threadhijacking.exe</code>	TEH Injection Sample
<code>C:\injection\apc\apcinjection.exe</code>	APC Injection
<code>C:\injection\mapview\mapview_remote.exe</code>	Map View Injection
<code>C:\injection\pe\peinjection.exe</code>	PE Injection
<code>C:\injection\hollow\hollow.exe</code>	Process Hollowing tool
<code>C:\injection\hollow\demo\Hollow.exe</code>	Hollow Sample

Tools Usage

Command	Description
<code>C:\injection\tools\Moneta64.exe -m ioc -p 1234</code>	Moneta - Enumerate only regions related to suspicious memory
<code>C:\injection\tools\pe-sieve64.exe /shellc 3 /pid 1234</code>	PE-Sieve - Scan the process memory for signs of a shellcode
<code>Invoke-AtomicTest T1055.002 -ShowDetails</code>	Check the details of a specific MITRE technique eg. T1055.002
<code>Invoke-AtomicTest T1055.002</code>	Execute Atomic Test with ID T1055.002
<code>msfvenom --platform windows --arch x64 -p windows/x64/exec CMD=calc.exe -e x64/xor -f c -v shellcode</code>	Generate msfvenom Shellcode to execute calculator app
<code>x86_64-w64-mingw32-gcc <program.c> -o <output.exe> -m64 -s</code>	Compile a C program into exe file
<code>x86_64-w64-mingw32-gcc -shared -o <output.dll> <program.c> -m64 -s</code>	Compile a c program into DLL file

WinDbg Commands

Command	Description
<code>.hh</code>	Opens the debugger help documentation
<code>dt [-rN] [type] <addr></code>	Displays type information for a symbol or structure at the specified address; -rN for recursion depth.
<code>dt _PEB <address></code>	Display details of the PEB structure
<code>dt _EPROCESS <address></code>	Display details of the EPROCESS structure

Command	Description
!peb	Show the Process Environment Block (PEB)
!teb	Show the Thread Environment Block (TEB)
!process 0 0	List all active processes
??	Evaluates and displays a C++ expression.
? [-rN] <expr>	Evaluates and displays a MASM expression; -rN for recursion depth.
poi(<addr>)	Dereferences and displays the pointer at the specified address.
!m	List all loaded modules in the process
u <address>	Disassemble instructions at a given memory address
bp <address>	Set a breakpoint at a memory address
!l	List all active breakpoints

WINAPI/NTAPI Functions Reference

Command	Description
OpenProcess	Get a handle to a target process
VirtualAllocEx	Allocate memory in a remote process
WriteProcessMemory	Write shellcode to the remote process
CreateRemoteThread	Create a thread in a remote process
QueueUserAPC	Inject an asynchronous procedure call (APC)
NtQueueApcThread	NTAPI variant for queuing an APC
NtAllocateVirtualMemory	Allocate memory using NTAPI

Command	Description
NtWriteVirtualMemory	Write memory using NTAPI
NtProtectVirtualMemory	Change memory protection of a region
NtCreateThreadEx	Create a remote thread using NTAPI
SetThreadContext	Modify thread execution context (used in thread hijacking)
SuspendThread	Pause a thread in a remote process
ResumeThread	Resume execution of a suspended thread
ZwUnmapViewOfSection	Unmap memory from a process (used in Process Hollowing)

Windows Security Event IDs Reference

Event ID	Description	Detection Use Case
4688	Process Creation	Detect suspicious processes
4656	Handle Requested for an Object	Detect OpenProcess
4657	Registry Modification	Look for persistence mechanisms
4663	Access Attempt on an Object	Track process injections
4697	Service Installation	Detect malware persistence
7034	Service Unexpectedly Terminated	Look for process crashes
7045	New Service Installed	Detect unauthorized services
6005	Event Log Service Started	Identify system restarts
6006	Event Log Service Stopped	Detect suspicious shutdowns

Sysmon Event IDs for Detection

Event ID	Description
Sysmon ID 1	Process creation
Sysmon ID 3	Network connection detected
Sysmon ID 7	Image loaded
Sysmon ID 8	CreateRemoteThread detected
Sysmon ID 10	Process access via OpenProcess
Sysmon ID 11	File creation time change
Sysmon ID 13	Registry key modification
Sysmon ID 17	Pipe creation

Tools Location in the Target (VM)

Name	Path
API Monitor	C:\Program Files\rohitab.com\API Monitor\apimonitor-x64.exe
x64dbg	C:\Tools\x64dbg\release\x64\x64dbg.exe
SysinternalsSuite	C:\Tools\SysinternalsSuite
ProcessHacker	C:\Tools\ProcessHacker\ProcessHacker.exe
Moneta	C:\Tools\Moneta64.exe
PE-Sieve	C:\Tools\pe-sieve64.exe
Atomic Red Team	C:\AtomicRedTeam\invoke-atomicredteam